Name: _____

YEAR 11 SPECIALIST MATHS INVESTIGATIVE TASK 2

Combinatorics: Pascal's triangle

Prerequisites

- Basic understanding of combinatorics, ${}^nC_r$
- The multiplication and addition principle for combinatorics
- Algebra skills
- Basic spreadsheet skills

[Total: 70 marks]

1 Pascal's triangle is shown in the diagram below.

1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

1 5 10 10 5 1

a Describe in words the rule for producing the numbers in Pascal's triangle.

b Copy the above and use this rule to complete the next two lines.

c Expand $(x + a)^2$, $(x + a)^3$, $(x + a)^4$ and $(x + a)^5$. You may use CAS.

How do these relate to the numbers in Pascal's triangle?

Predict the expansion of $(x + a)^7$. Write down your answer.

d Consider the expansion of $(x + a)^7$.

$(x + a)^7 = (x + a)(x + a)(x + a)(x + a)(x + a)(x + a)(x + a)$

Using selections of xs, how many times will you come up with a^3x_4? Express your answer in the form ${}^nC_r$.

e Consider the expansion of the binomial $(x + a)^n$.

Using selections of xs, how many times will you come up with $an - rxr$? Express your answer in the same form as (**d**).

f Suppose the expansion of $(x + a)^n$ is $(x^n + \ldots + Aa^{n-r}x^{r+1} + Ba^{n-r+1}x^r + \ldots + a^n)$.

By first expressing $(x + a)^{n+1}$ as $(x + a)(x + a)^n$, find the coefficient of $a^{n-r}x^{r+1}$ in the expansion of $(x + a)^{n+1}$, using the above expansion.

How does this relate to Pascal's triangle?

© Cengage Learning Australia Pty Ltd 2013

g Hence write the first six lines of Pascal's triangle as combinations. Also write the expansion of $(x + a)^5$ in terms of combinations, and the first four terms of $(x + a)^n$ in terms of combinations.

How would you explain the top number (1) as ${}^0C_0$ conceptually? How does it fit the factorials in the formula for ${}^nC_r$?

h i A new set is made by selecting objects from a set containing n objects. By considering that you could choose either no objects or one object or two objects etc., find a rule for the number of ways the new set can be formed, in terms of a sum of combinations.

ii By letting $x = a = 1$, use the expansion of $(x + a)^n$ to find a simpler expression for your expression in part **i**.

iii Confirm this rule by adding across rows of Pascal's triangle.

iv Consider another way of making the new set. You can considering the new set as a sequence of choices whether or not to include each object, use the multiplication rule to show that:

$$ {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots + {}^nC_n = 2^n. $$

[2 + 2 + 4 + 1 + 1 + 4 + 4 + (2 + 3 + 1 + 2 = 8) = 26 marks]

2 Use a spreadsheet to investigate Pascal's triangle. The spreadsheet will produce the version of Pascal's triangle below. We'll start with 11 rows.

1

1 1

1 2 1

1 3 3 1

- - - -

Insert 1 in cell A1.

The first row will be all 1s, so drag A1 down to A11.

In B2, insert the formula = A1+ B1.

Drag this down to B11.

Highlight from B2 to B11. Drag from the bottom cell across to the right until you get a 1.
This should take you as far as column K.

You may find it easier to reduce column width. Click on  (top left) to highlight the whole spreadsheet.
Choose Format and Column Width and select an appropriate width. About 5−6 is good.
If using Excel 2010 you must be in the Home menu.

a Create this spreadsheet. You may wish to check it with your teacher.

b Find the sum of each line of Pascal's triangle

In the top cell of the first blank column on the right of Pascal's triangle enter

= SUM(

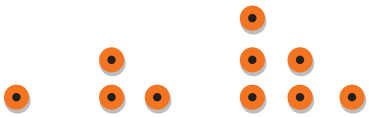
© Cengage Learning Australia Pty Ltd 2013

Highlight that row of Pascal's triangle and press Enter. You should get 1 in the cell.

Drag this down till you reach the bottom row of Pascal's triangle. What do you observe about the pattern in the sums of the rows?

Check your results with your teacher.

c The triangular numbers are 1, 3, 6, 10, ….



Find the triangle numbers in Pascal's triangle.

Use what you have discovered about Pascal's triangle to find a formula for the rth triangle number.

d What do you notice if you add two consecutive triangle numbers?

Prove this algebraically and verify it on the spreadsheet.

Note: After entering one sum, you can drag, or copy and paste, to get other sums on the spreadsheet.

You can enlarge the triangle by dragging all columns with numbers down, then dragging the right row of numbers across.

e Enlarge your triangle to go at least to row 15 (starting 1, 15, …).

Use this enlarged triangle to evaluate ${}^{14}C_{9.}$ In which cell is it?

f Refer to the hockey stick numbers in **Exercise 5.3**, question **7**. The top of a hockey stick number is always a '1'.

What shapes do the hockey stick numbers make on the spreadsheet? Verify the hockey stick rule works with some large numbers.

g Refer to the 'Star of David' numbers in *Exercise 5.3*.

What shapes do these numbers make on the spreadsheet? Verify the rule works with some large numbers. Can you use the drag down (or across) method to verify neighboring stars?

[5 + 2 + 6 + 6 + 3 + 5 + 5 = 32 marks]

3 a Why would you expect ${}^{n}C_{r}$ to equal ${}^{n}C_{n-r}$? Prove this algebraically.

b Hence show that if a row contains an even number of terms, the sum of the odd terms in a row will equal the sum of the even terms in a row.

c Use your spreadsheet to check out the cases when there are an odd number of terms. You will find the drag down facility useful.

d If you take the terms three at a time, and compare the sum of the 1st, 4th, 7th, … terms with the sum of the 2nd, 5th, 8th, … terms and the sum of the 3rd, 6th, 9th, … terms, why can't they all be equal?

e Use your spreadsheet of Pascal's triangle to look for any patterns or rules in part **d**.

[3 + 4 + 1 + 1 + 3 = 12 or more marks]

© Cengage Learning Australia Pty Ltd 2013